

Decision Support System (DSS)

1 What is DSS?

DSS Definition

A **Decision Support System (DSS)** is a computer program used to improve a company's decision-making capabilities. It analyzes large amounts of data and presents an organization with the best possible options available.

DSSes bring together data and knowledge from different areas and sources to provide users with information beyond the usual reports and summaries. This is intended to help organizations make **informed decisions**.

1.1 Typical Information Gathered

A decision support application might gather and present the following typical information:

- **Comparative sales figures** between one week and the next.
- **Projected revenue figures** based on new product sales assumptions.
- **The consequences** of different decisions.

2 Advantages and Disadvantages

2.1 Advantages

- **Informed Decisions:** Facilitates better, up-to-date and informed decisions by considering multiple sources.
- **Different Outcomes:** Considers various business outcomes based on historical data.
- **Efficiency:** Automates analysis of large data sets.
- **Collaboration:** Often includes communication and collaboration features.
- **Flexibility:** Usable across many different industries.
- **Handle Complexity:** Manages problems with multiple dependencies.

2.2 Disadvantages

- **Cost:** High expenses for development, implementation, and maintenance.
- **Dependence:** Risk of over-reliance, reducing human subjectivity.
- **Complexity:** Requires vast data and complex design.
- **Security:** Sensitive data requires increased focus on security.
- **Resistance:** Employees may resist machine-based workflow changes.

3 Data-Driven Decision Making (DDDM) Examples

Real-World Applications

- **Starbucks (Site Selection):** Starbucks uses geographic information systems (GIS) to analyze demographic data and traffic patterns to determine the most profitable locations for new stores.
- **Netflix (Content Strategy):** Netflix analyzes viewer behavior, search history, and ratings for over 150 million subscribers to decide which shows to produce (e.g., *House of Cards*) and provide personalized recommendations.
- **Amazon (Supply Chain):** By leveraging big data and AI, Amazon predicts consumer trends and optimizes inventory management, resulting in a significant increase in average order value and reduced shipping times.

DSS transforms raw data into actionable insights, driving organizational growth and competitive advantage.